

Chapter 9

Name: _____

1. Define the following:

- a. Pasteurization _____

- b. disease _____

- c. microbes _____

- d. pathogens _____

- e. bacteria _____

- f. viruses _____

- g. fungi _____

- h. protist _____

- i. parasites _____

- j. communicable _____

- k. non-communicable _____

- l. botulism _____

- m. salmonellosis _____

- n. sanitary _____

- o. poverty _____

- p. staph bacteria _____

- q. sterilization _____

- r. aseptic _____

2. Describe 5 factors that may interact with one another and possibly make you sick.
Pg 158

3. What did the following scientists discover.

a. Robert Koch pg 158

b. Louis Pasteur pg 158

c. Florence Nightingale pg 170

d. Joseph Lister pg 170

4. List the conditions that bacteria need to grow. Explain why each is important. Pg 159

5. What is a pathogen? Give three examples of pathogens, pg 159 to 162

6. Explain how viruses such as the AIDS virus spread. pg 159
7. How do protists spread? Pg 162
8. What causes beaver fever? Pg 162
9. What causes athletes foot? Pg 162
10. What might happen to the bacteria in a hamburger you left on the counter for a few hours? Explain why. Pg 159

11. List 3 types of communicable disease. Pg 163

12. List 3 types of non-communicable disease. Pg 163

13. How does a clean environment protect you from disease? Pg 164

14. List four factors that affect the risk of developing a non-communicable disease.
Pg 165

15. Complete the following table.

Disease a family member has gotten	Communicable or non-communicable	Effects of each disease

16. What is poverty and how does it contribute to the spread of disease? Pg 166

17. Use Table 9.2 to identify what diseases you may be at risk of getting. Explain how you could reduce your risk of getting each one. Pg 167

18. What types of disease kills most people in poor countries? In rich countries? Explain why you think this is so. Pg 167

19. Explain how each of the following methods of food preservation protects food from pathogens: vacuum packaging, canning, freezing and drying. Pg 168

20. List 5 recommendations for protecting food preparation surfaces and utensils from contamination. Pg 169

21. Explain why you would put an antiseptic and a disinfectant in a first aid kit. Pg 170

22. What is the purpose of sterilization? Pg 170

23. List 4 aseptic practices. Pg 170

1. Define the following:a) Pandemics _____

_____b) Epidemics _____

_____c) Quarantine _____

_____d) Public Health _____

_____e) Vaccinations _____

_____f) Migration (pg184) _____

_____g) AIDS (pg 179) _____

_____h) Tuberculosis (pg 178) _____

_____i) Bubonic Plague (pg 178) _____

_____j) Advisory (pg 185) _____

2. Who is responsible for the preserving the quality of water you use and drink? Pg 174
3. Why was smallpox tuberculosis and the flu so deadly to First Nations Peoples? Pg 176
4. What were the effects of disease on Aboriginal Groups? Pg 176
5. How is a pandemic different from other diseases? Give an example of a pandemic disease. Pg 176
6. When does an epidemic become a pandemic? Pg 178

7. Complete the following table:

	Bubonic Plague	Tuberculosis	Spanish Flu	AIDS
Social Conditions				
Impact of the pandemic				
When and Where the disease was present				

8. Why did society begin to take action to protect the health of the community? Pg 180

9. What jobs are the provincial health department responsible for? Pg 181

10. What does the Regional health authority do when it comes to water quality? Pg 181

11. How do landfills prevent toxic chemicals from entering the water system? Pg 181

12. Why do we treat human and animal waste? Pg 182

13. If the air quality is between 50 to 100 (Poor to Very Poor) what kind of alert or warning do you think would be sent out? Pg 182

14. List 6 vaccinations that are given by the Health Authority. Pg 182

15. List the 4 points put forward by the BAC campaign. Pg 183

16. Complete the following Table pg 184

Factor	Concern	Ways to reduce risk
Population growth and overcrowding		- disease monitoring - Emergency response plans to deal with a community with disease
Migration	Diseases may be carried from country to country or from rural to urban	
	People eat out in restaurants	- restaurant inspections - sanitary work places
Travel		
Food supply		- Food Inspectors check food supply - Transportation vehicles are refrigerated
Sexual activity and child birth		

17. Do you think it is important to have public health guidelines? Use examples to help you explain why or why not. Pg. 187

18. Read the Walkerton's Water Tragedy Timeline on page 186. Do you think that the PUC manager should face criminal charges? Explain why you think the way you do. If you think that he should be charged what do you think he should be charged with?

1. Define the following:

- a. Antibodies _____

- b. Physical defenses _____

- c. Inflammatory response _____

- d. Immune system _____

- e. Macrophages _____

- f. Antigens _____

- g. Inherited Immunity _____

- h. Acquired Immunity _____

- i. Active Immunity _____

- j. Passive Immunity _____

- k. Vaccine _____

l. Immunization _____

m. Over-the-Counter Drugs _____

n. Antibiotic _____

2. Where does penicillin come from? Pg 192

3. What is the first line of defense against infection? Pg 194

4. Complete the Chart on Physical Defenses Pg 194

Structure	Function
	Contain hair and mucous, which trap pathogens before they can enter the respiratory system Warm incoming air
	Catch pathogens before they enter the body. May become infected to help the immune system resist and fight future infections Store White Blood Cells
Eyelids	
Tears	
	Protect the Eye
Hair	
	Contains wax which traps pathogens and particles
	Consists of a thick barrier with an acidic surface that prevents pathogens from entering Prevents loss of body fluids
Stomach	

5. Why do smokers have a nagging cough? Pg 194

6. What is the 2nd line of defense? Pg 195

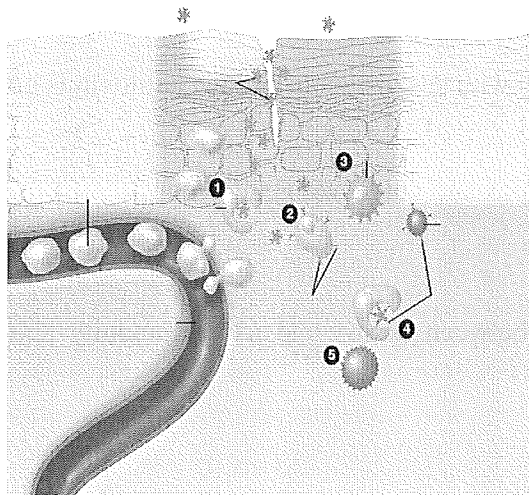
7. List 3 different things that can trigger an inflammatory response. Pg 195
8. What is happening when your skin swells, turns red, and gets warm around a sliver? Pg 195
9. What do macrophages destroy? Pg 195
10. What is the 3rd line of defense? Pg 195
11. Read the following scenarios and write whether the following have antigens. Pg 197
 - a. A virus enters the blood stream.
 - b. A red blood cell travels from the heart to the lungs
 - c. Bacteria transferred from a piece of broken glass enters the body.
 - d. Cancer cells in the body replicate.
 - e. A macrophage consumes a damaged cell.

12. What does your body release when it senses an antigen? Pg 197

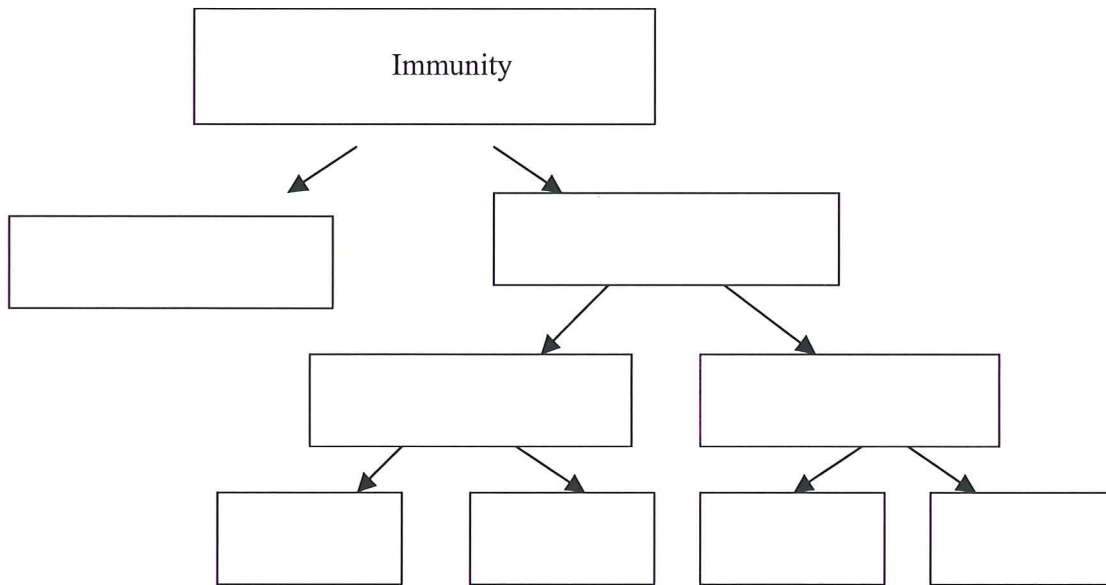
13. How do antibodies work to give us immunity? Pg 197

14. Look at the following diagram. Order the following in proper chronological order from 1 to 5. Pg 198

- _____ a. Antigens from dead bacteria stick out of the macrophage
- _____ b. Antibodies produced by B-cells attach to antigens and Killer T-Cells destroy them.
- _____ c. Bacteria enter the body and macrophages surround them
- _____ d. B cells produce memory cells to store the information about how to kill the antigen so they can make the antibodies faster the next time.
- _____ e. Helper T cells read the antigens and B-cells make and release antibodies



15. Complete the following flowchart. Pg 199



16. Classify the following as a type of immunity. Pg 199

- a. A baby gets immunity from diseases from a mother's milk.
- b. You are vaccinated against small pox.
- c. A cow is injected with snake venom and the antitoxin is given to you.
- d. You get chickenpox and you never get it again.

17. What is the difference between passive and active immunity? Give an example of each. Pg 199

18. Why do people with A blood reject transplants from people with B blood? Pg 200

19. Why are people with O negative blood considered universal donors? Pg 200

20. What is the difference between an injection and a vaccine? Pg 201

21. How do vaccines work? Pg 201

22. Why should you get immunizations? Do you think some countries are justified in not giving them? Pg 202

23. How is your immune system related to allergies? Pg 202

24. What is the difference between over-the-counter drugs and antibiotics? Pg 204

25. Why do antibiotics not work on colds? Pg 204

26. List 3 problems with antibiotics. Pg 205

27. List 3 things you should do to use antibiotic safely? Pg 206

28. What should you do when choosing Over-The-Counter Drugs? Pg 206

Chapter 12 Genetics and Health

Name: _____

1. Define the following:

a) Genetics _____

b) Chromosomes _____

c) DNA _____

d) Genes _____

e) Heredity _____

f) Dominant _____

g) Hybrid _____

h) Purebred _____

i) Recessive _____

j) Genotype _____

k) Punnett Square _____

l) Pedigree _____

m) Mutation _____

n) Mutagens _____

o) Genetic Disorders _____

p) Genetic Engineering _____

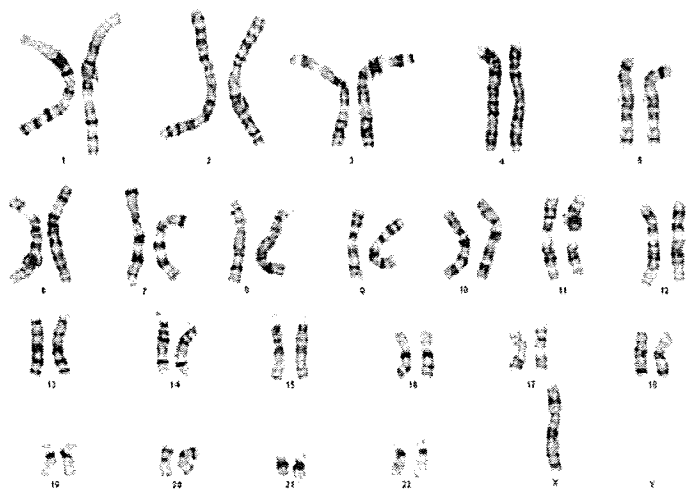
q) Amniocentesis _____

2. Where in the cell are the chromosomes? Pg 212

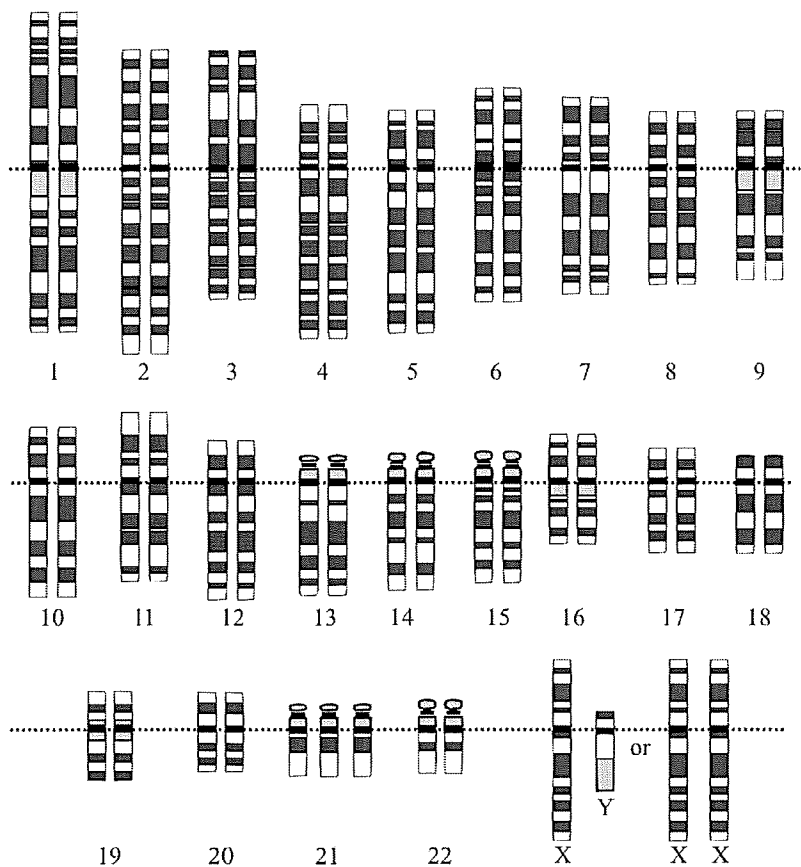
3. If one side of the DNA molecule looks like this A-G-C-T-G What must the other side look like? Pg 212

4. How many chromosomes in a normal human? Pg 213

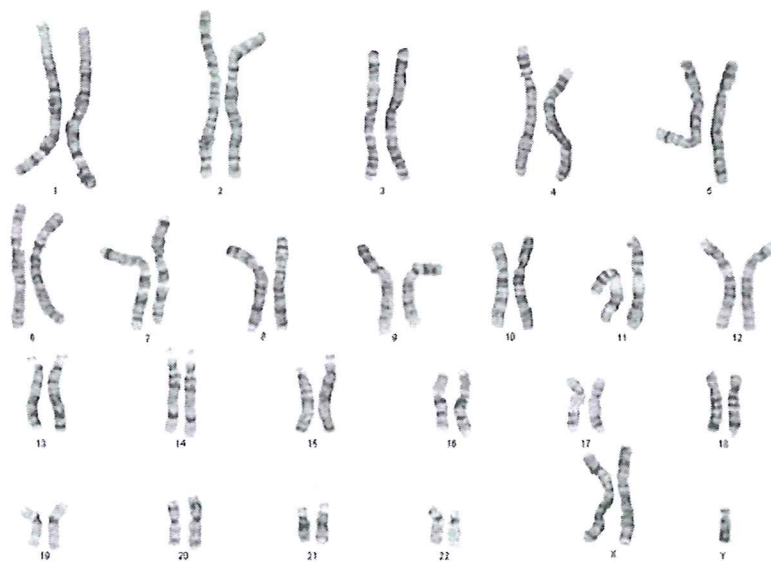
5. Here are 3 genetic disorders Identify what the person has using the following karyotypes. Downsyndrome = Trisomy 21 (3chromosomes), Turners Syndrome = a single X, Klinefelters Syndrom XXY Pg 213



Disorder: _____



Disorder: _____



Disorder: _____

6. For the following state whether the person is male or female. Pg 213

a)XX

b)XY

7. Why can it be said that the male determines the sex of the baby? Pg 213

8. Who studied pea plants and became known as the father of genetics? Pg 216

Use the following Information to answer the next two questions.

TT or tt = Homozygous (two of the SAME genes)

Tt = Heterozygous (two different genes)

9. Make a Punnett Square using the following data. Tall Plants are Dominant to Short Plants. A heterozygous tall plant mates with a short plant. Pg 217

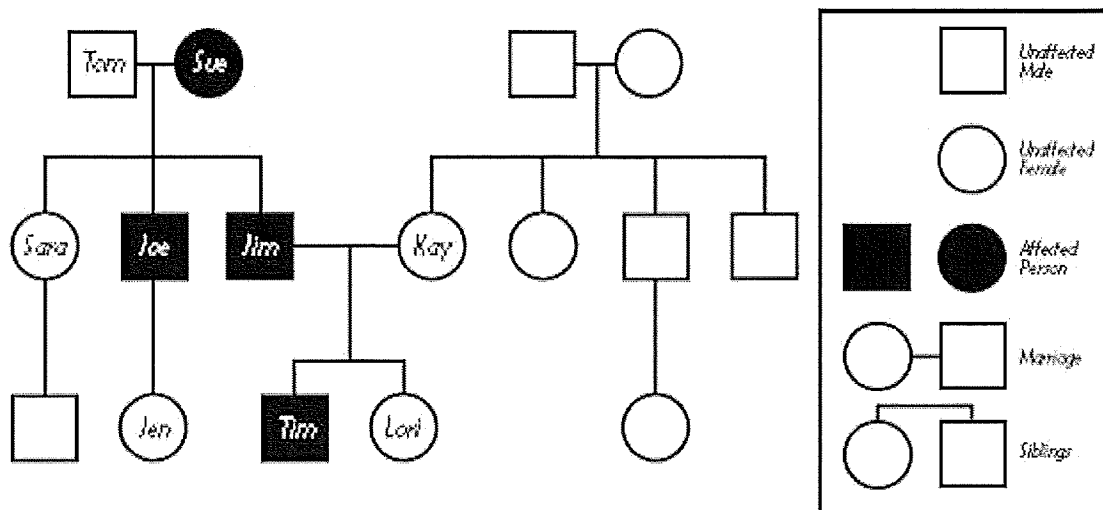
10. Brown Eyes are Dominant to Blue Eyes. If a Father who is homozygous for Brown Eyes mates with a woman with Blue Eyes what chance does the offspring have of having blue eyes? Pg 217

11. Draw the punnett square for the following situation and fill out the chart below. In sheep black wool is recessive to white. So two white heterozygous sheep mate. Pg 217

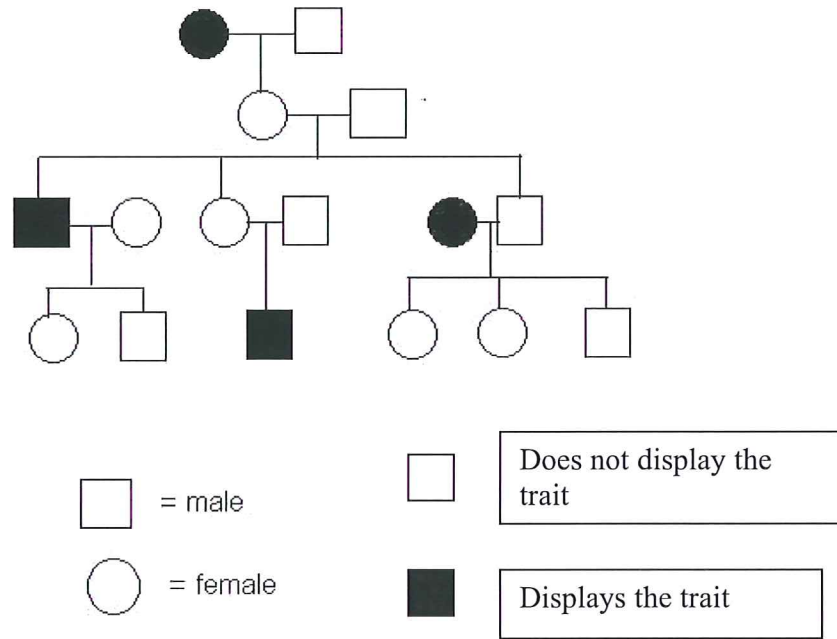
	Genotype (The Genes)	Phenotype (What it looks like)	Percentage
Offspring 1			
Offspring 2			
Offspring 3			
Offspring 4			

What is the ratio of White sheep to Black Sheep?

12. Look at the following pedigree. And answer the following questions. Pg 218



- How many generations does this pedigree go?
- Name the infected people.
- How many children does Jim have?



13. Using the pedigree above determine if the infected trait is dominant or recessive.
 EXPLAIN why. Pg 218

14. Construct a pedigree using the following information.

1. Brown eyes is dominant to Blue eyes
2. Mr. Pederson is homozygous for brown eyes marries a Blue eyed Girl
3. They have 3 children, 2 sons and a daughter. They all have brown eyes.
4. One of the sons and the daughter do not have children.
5. 1 son marries a brown eyed girl. They have 2 sons and daughter. Both sons have blue eyes.
6. Be sure to include a key. Pg 218

Key	
☐	
☐	
☐	
☐	

15. Describe how a mutation might occur when chromosomes are copied. Pg 220

16. a) What are mutagens? PG 220

b) Name one example of a mutagen and a genetic disorder that results from it. Pg 222

17. Why should pregnant women not drink or do drugs? Pg 222

18. Why might amniocentesis pose an ethical dilemma for some people? Pg 223

19. Complete the following chart on advances in genetic research. Pg 223 and 225

Pros of Genetic Research	Cons of Genetic Research

20. Should they allow cloning of people so people may harvest organs from their clone in the case of a medical emergency?

